frontiers in Pediatrics

TYPE Original Research
PUBLISHED 04 November 2026
DOI 10.3389/fped.2026.1897283

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OPEN ACCESS

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RECEIVED 01 June 2026
REVISED 08 September 2026
ACCEPTED 16 October 2026
PUBLISHED 04 November 2026

CITATION
Hu J, Liu H, Chen Y, Zhang W, Yang Z, Yang B and Yin L (2026) Development and psychometric evaluation of the Chinese school football transition social support scale among adolescents: evidence from a predominantly central China sample.
Front. Pediatr. 14:1897283.
doi: 10.3389/fped.2026.1897283

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Development and psychometric evaluation of the Chinese school football transition social support scale among adolescents: evidence from a predominantly central China sample

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Background: Social support received by adolescent student-athletes may influence whether they continue participating in sport. However, the existing evaluation tools mainly focus on interpersonal relationships such as family, coaches and peers. In China's school-football system, adolescents' participation is also shaped by institutional conditions, and existing instruments may be insufficient for assessing perceived support. Therefore, there is a need to develop a context-sensitive assessment tool.

Methods: A total of 541 adolescents were included, of whom 93.2% were recruited from Central China. Ecological systems theory guided the conceptual organization of the five social-support sources assessed by the scale. Scale items were generated based on a literature review, expert reviews, and student cognitive interviews. Exploratory Factor Analysis and Confirmatory Factor Analysis were utilized within a multi-stage cross-validation design to evaluate the scale's structure. Test-retest reliability and measurement invariance were examined to confirm stability and equivalence. Furthermore, known-groups comparisons, logistic regression, and receiver operating characteristic (ROC) analysis were conducted to assess the scale's capacity to distinguish between continuing and dropout groups.

Results: The finalized Chinese School Football Transition Social Support Scale (C-SSST-SF) consists of 26 items capturing five distinct dimensions: governmental, school, family, coach, and peer support. Confirmatory factor analysis indicated acceptable fit for the five-factor model (χ^2/df = 1.62, CFI = 0.936, TLI = 0.928, RMSEA = 0.048, SRMR = 0.050). The scale demonstrated strong internal consistency (Cronbach's α = 0.828–0.923) and stable 2-week test-retest reliability. Furthermore, after establishing scalar invariance, results indicated that dropouts perceived significantly less support across all dimensions compared to their continuing peers.

Conclusions: The C-SSST-SF demonstrated satisfactory psychometric properties in the present sample and may provide a context-sensitive measure of perceived interpersonal and institutional support during the school-football transition. Supplementary classification analyses provided preliminary evidence of differences by participation status, but the limited sensitivity does not support use of the scale as a standalone screening tool. Broader geographic validation is needed before its applicability across Chinese school-football populations can be established.

KEYWORDS

adolescent athletes, Chinese school football, measurement invariance, psychometric evaluation, scale development, social support

1 Introduction

Organized sport participation of adolescents is recognized as a key way to promote physical and mental health and social adaptation (1–3). However, dropout from organized sport remains common during adolescence and continues to be a concern in youth sport research (4, 5). In this context, dropout refers to a sustained cessation of school-football participation rather than a temporary interruption. Social support has been widely used to explain individual differences in sport continuation and dropout (6, 7). Previous studies have pointed out parents, coaches, and peers as important sources of support, particularly in relation to motivation, perceived competence, and belonging (8, 9). For this reason, research on sport continuation and dropout depends not only on recognizing social support as a multidimensional construct, but also on using instruments that reflect the sources of support that matter in a given sport and educational context.

Existing measures have made important contributions to the study of social support in sport. The Multidimensional Scale of Perceived Social Support assesses perceived support from family, friends, and significant others (10). There are also sports-specific PASS-Q (11) and ARSQ (12) to evaluate the function and actual frequency of support received, including tools such as PISQ (13) and CART-Q (14) for parental support and coach-athlete relationship. These past scales and literature have confirmed that interpersonal interaction from micro systems—that is, the support of families, peers and coaches—is the core driving force for maintaining youth sports participation (5, 15). However, these tools originating from the Western context rely on the underlying assumption of "open resources", that is, assuming that as long as individuals have the will and receive interpersonal encouragement, the social environment will naturally provide opportunities to participate (16). However, this assumption may not fully apply to the Chinese school-football context (17).

As one of the most populous countries in the world, China has implemented a unique "School Football" national strategy to cope with the double crisis of declining physical fitness among youth and lack of football reserve talents (18, 19). Unlike independent club structures, China's youth football development is deeply anchored within the public education system and steered by national policy (20). There may be a problem in the development of this model: If there is a lack of guarantees at the macro institutional level—such as the school's course scheduling permission (21, 22) and government-backed educational advancement pathways—simple interpersonal emotional support is often difficult to translate into practical participation (23, 24). Sustaining involvement requires navigating school-specific barriers like academic schedules, facility limits, and competition access (25). These hurdles culminate during the critical transition into junior high school (26). As academic demands intensify during this transition, conflicts between academic study and football training and discontinuities in developmental pathways may contribute to disengagement from football participation (27, 28). These contextual features suggest that social support in school football should be measured across both interpersonal and institutional sources.

However, existing scales generally overlook schools as organizational carriers and governmental policies as institutional guarantees. Therefore, the present study argues that a cross-level framework is needed to integrate interpersonal and institutional sources of support. For this reason, we introduce Bronfenbrenner's ecological systems theory (29) to expand the concept of micro-social support. In this integrated model, support from family, peers, and coaches is conceptualized as "proximal interpersonal support" that is directly experienced by adolescents. Given football's high degree of tactical interdependence and group collaboration (3), peer acceptance and the quality of coach interactions within the microsystem may provide important emotional feedback and help sustain adolescents' sense of belonging and motivation for continued participation (30). School support refers to organizational support directly experienced within the school, whereas governmental support refers to broader policy- and institutional-level support that shapes participation opportunities beyond the individual school (31).

Abbreviations

AUC, area under the curve; AVE, average variance extracted; CFA, confirmatory factor analysis; CFI, comparative fit index; CR, composite reliability; C-SSST-SF, Chinese School Football Transition Social Support Scale; EFA, exploratory factor analysis; HTMT, Heterotrait–monotrait ratio; ICC, intraclass correlation coefficient; I-CVI, item-level content validity index; MDC, minimal detectable change; MGCFA, multi-group confirmatory factor analysis; OR, odds ratio; RMSEA, root mean square error of approximation; ROC, receiver operating characteristic; S-CVI/Ave, scale-level content validity index, average; SEM, standard error of measurement; SRMR, standardized root mean square residual; TLI, Tucker–Lewis index.

Based on this gap, the present study followed DeVellis's recommendations for scale development to develop the Chinese School Football Transition Social Support Scale (C-SSST-SF) (32). The scale was designed for adolescents involved in Chinese school football during the transition from primary to junior secondary school. Social support was the focal construct, and ecological systems theory was used to organize the sources of support that adolescents may perceive in school football. The scale included five sources of support: family, coach, peer, school, and governmental support. The study examined initial evidence for the scale's reliability and validity through item development, expert review, cognitive interviews, exploratory and confirmatory factor analyses, reliability estimation, measurement invariance testing, known-groups comparisons, and supplementary classification analyses using logistic regression and ROC analysis. It was hypothesized that adolescents who continued participating in school football would report higher overall and dimension-specific social support than those who had dropped out.

2 Materials and methods

2.1 Participants and setting

Questionnaires were administered online and distributed through the researchers' personal contacts with students and teachers, who helped forward the survey to adolescents with current or previous school-football experience across China. A total of 541 valid responses were obtained. Participants were not recruited separately according to participation status; dropout status was subsequently determined based on self-reported training history and current participation status according to the predefined criteria described below.

According to participants' current football participation behavior, the study divided them into two groups: (1) continuing group: defined as students who have a systematic training background for at least 1 year and are still participating in school team practices or competitions ($n = 419$, 77.4%); (2) dropout group: defined as students who have received systematic training for at least 1 year but have stopped participating in any form of school football training or formal competition for more than 6 consecutive months at the time of the survey ($n = 122$, 22.6%). The setting of this time threshold is based on the time structure of the Transtheoretical Model on behavioural change (33, 34). Individuals who stopped training because of severe physical injuries, such as fractures or ligament tears, were excluded. This exclusion criterion was used to focus on non-injury-related dropout that may be more closely associated with social, educational, or environmental factors.

The study was approved by the Biomedical Research Ethics Committee of Hunan Normal University (Approval Number: 2026–306). Participation was voluntary, and informed consent was obtained from parents or legal guardians, and assent was obtained from adolescent participants before data collection. The demographic characteristics, geographic distribution, and football participation status of the final sample are summarized in Table 1.

TABLE 1 Demographic characteristics and football participation status of the sample.

Characteristics	Category	*n*	%
Gender	Male	406	75.1
	Female	135	24.9
Grade level	Primary school		
	Grade 5	122	22.6
	Grade 6	104	19.2
	Junior high school		
	Grade 7	95	17.6
	Grade 8	134	24.8
	Grade 9	86	15.9
Geographic region	Central China (e.g., Hunan, Henan)	504	93.2
	Eastern China (e.g., Guangdong, Jiangsu)	35	6.5
	Western China (e.g., Guizhou)	2	0.4
School football participation status	Continuing group	419	77.4
	Dropout group	122	22.6

2.2 Scale development

The development of the C-SSST-SF followed DeVellis's recommendations for scale development, including construct definition, item generation, content review, and empirical testing (32). The scale was designed to assess adolescents' perceived support for continued participation in Chinese school football during the transition from primary to junior secondary school. Ecological systems theory was used to organize the sources of support that adolescents may perceive in school football. Five sources of support were specified: governmental, school, family, coach, and peer support.

Family support items were mainly informed by PISQ (13) and PASS-Q (11), together with the Chinese version of the Child and Adolescent Social Support Scale (35). Coach support was informed by CART-Q (14) and previous research on coach autonomy support and youth soccer dropout (36). At the same time, context-specific items were added to reflect China's sport–education integration context, such as coaches' concern for students' academic performance. Peer support was adapted from PASS-Q (11) and QRA (37), with additional high-involvement behaviors such as post-game joint summary and accompanying extra practice. Items for school and governmental support were generated from the Chinese school football context, relevant policy documents, and prior research on school sport participation environments (19, 38). School support items explicitly referred to the school and school personnel, whereas governmental-support items used source-specific cues such as government, official channels, and policy-related pathways to cue the intended source distinction.

The research team first generated 56 candidate items. After internal review for redundancy, clarity, and age appropriateness, 36 items were retained. A panel of 19 physical education scholars and experienced school football coaches then completed

two rounds of Delphi review. In the first round, experts evaluated each item in terms of dimensional fit, relevance, clarity, and age appropriateness. In the second round, experts rated the importance of the remaining items and suggested wording changes (39). Items were revised, merged, or removed according to expert ratings and written comments. Experts rated item relevance, clarity, dimensional fit, and importance using a 5-point scale. Item-level content validity index I-CVI, scale-level content validity index S-CVI/Ave, coefficient of variation, and modified kappa were calculated to evaluate expert agreement and chance-corrected content validity (40, 41). The content validity index and expert review results are provided in Supplementary Table S1.

Ten school football participants from upper primary and junior secondary grades then completed cognitive interviews using a think-aloud procedure. The purpose of these interviews was to identify ambiguous terms and revise items into language that adolescents could understand (42). After expert review and cognitive interviews, 30 items were retained for pilot psychometric testing. The complete item descriptions for the initial 30-item version are provided in Supplementary Table S2, and the full process of item generation, content validation, pretesting, and psychometric evaluation is summarized in Figure 1.

2.3 Measure

The 30-item pilot version of the C-SSST-SF was administered using a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree. Following the initial item analysis and EFA, we refined the instrument down to 26 items. These remaining items map distinctly onto five sources of support: governmental, school, family, coach, and peer support. To calculate the scores, responses were averaged within each specific dimension, and an overall composite score was calculated as the mean across all 26 items, with higher scores indicating stronger perceived support.

2.4 Procedure

All survey procedures were conducted online. Participants were recruited through the researchers' personal contacts with students and teachers, who helped forward the online survey to students in Grades 5–9 with current or previous school-football experience. Ethical safeguards were established immediately: participants had to read a brief overview of the study before seeing a single demographic question or scale item. We made it unequivocally clear that taking part was entirely optional and completely confidential. Finally, a convenience subsample of 53 participants completed the same scale again after a 2-week interval to evaluate temporal stability.

2.5 Data analysis

All analyses were conducted in R version 4.5.2 (43), with CFA and MGCFA performed using the lavaan package. The cleaned dataset was screened for missing values, invalid responses, and item distributions. The selection of measurement properties was informed by COSMIN terminology (44). Descriptive statistics, skewness, and kurtosis were calculated for all items (45). The full sample was randomly divided into an exploratory sample ($n = 270$) and a validation sample ($n = 271$) (46). Confirmatory factor analysis, internal consistency estimation, and convergent and discriminant validity testing were conducted in the validation sample. Item analysis and exploratory factor analysis were conducted in the exploratory sample. Corrected item-total correlations (CITCs) and extreme-group comparisons were used to examine item discrimination. A CITC value below .40 was considered indicative of a potentially problematic item. Items were considered for removal when they showed weak item-total correlations, poor discrimination, problematic wording, or unsatisfactory factor-analytic performance. Sampling adequacy for exploratory factor analysis was assessed using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's test of sphericity. After completing the initial item purification, principal axis factoring and Promax rotation were used for exploratory factor analysis to examine the underlying factor structure of the scale (47, 48).

Confirmatory factor analysis was conducted using the robust maximum likelihood estimator (MLR) because the item responses showed some departure from normality (45). Model fit was evaluated using χ^2/df, CFI, TLI, RMSEA, and SRMR (49), with the scaled χ^2 statistic and robust CFI, TLI, and RMSEA reported. Standardized factor loadings and average variance extracted were used to examine convergent validity (50). Values of AVE above .50 were considered acceptable. Discriminant validity was examined using latent factor correlations, the Fornell–Larcker criterion, and the heterotrait–monotrait ratio of correlations (50, 51). HTMT values below .85 were considered acceptable. A single-factor model was also tested as a competing structure to examine whether the item covariance could be adequately represented by one general factor. To further examine the relatively high correlations among family, coach, and peer support, two alternative factor structures were tested in the validation sample. A higher-order model specified the five first-order support dimensions under a general social support factor, whereas a three-factor model specified governmental support, school support, and a proximal interpersonal support factor combining the family, coach, and peer support items.

Measurement invariance across the continuing and dropout groups was tested using multi-group confirmatory factor analysis (MGCFA) with the same MLR estimator, sequentially comparing the configural, metric, and scalar invariance models. A decrease in CFI no greater than .010 was considered acceptable (51, 52). Internal consistency was examined using Cronbach's α coefficient and composite reliability (CR), with a criterion of >0.70 (50, 53). Based on the retest subsample ($n = 53$), the intraclass correlation coefficient (ICC) was used to evaluate the temporal stability over the 2-week interval using a criterion of >0.70 (54). Furthermore, in response to the COSMIN guidelines' emphasis on absolute measurement error, we calculated the standard error of measurement (SEM) and the minimal detectable change (MDC) to quantify the range of random score fluctuations and the threshold for true change (55).

Known-groups evidence was examined by comparing C-SSST-SF scores between the continuing and dropout groups. We

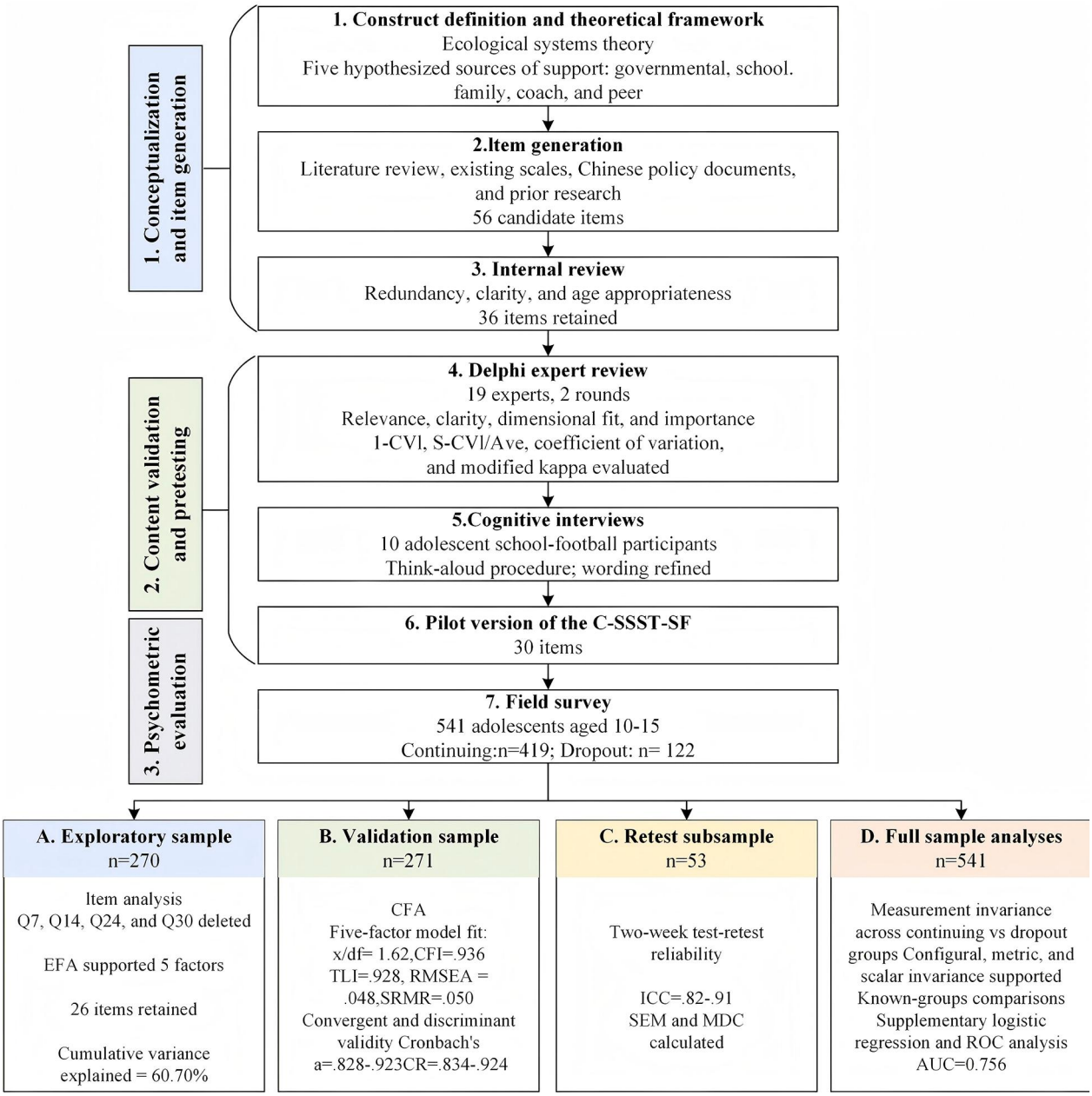


FIGURE 1
Flow chart of C-SSST-SF development and psychometric evaluation.

expected the continuing group to report higher perceived social support than the dropout group. Independent-samples *t* tests were conducted, and Cohen's *d* was calculated to estimate effect size (56).

To further examine whether C-SSST-SF scores were associated with school football participation status, supplementary classification analyses were conducted. Binary logistic regression was used with group membership coded as 0 = continuing and 1 = dropout. The total C-SSST-SF score was entered as the primary predictor, and odds ratios (ORs) with 95% confidence intervals were reported. Additional single-predictor logistic regression models were conducted for the five subscale scores separately because the dimensions were moderately to strongly correlated. Receiver operating characteristic analysis was then conducted to evaluate the ability of the total C-SSST-SF score to distinguish continuing participants from dropout participants (56, 57). The area under the curve (AUC), 95% confidence interval, optimal cut-off based on the Youden index, sensitivity, and specificity were reported. These analyses were interpreted as supplementary known-groups and classification evidence rather than criterion-related validity evidence.

3 Results

3.1 Content validity

The quantitative results of expert review (see Supplementary Table S1) show that the overall average content validity index (S-CVI/Ave) of the scale reaches 0.919, which exceeded the

recommended value of 0.90 (41). These results provided evidence supporting the content validity of the proposed item set.

At the item level, Q24 and Q30, both reverse-worded items, had marginal I-CVI values of 0.684, whereas all remaining items scored ≥0.78. At the content-validity stage, these two items were revised based on expert feedback and provisionally retained for subsequent empirical testing rather than deleted.

3.2 Item analysis

The results (Table 2) show that the corrected item-total correlations of Q7 ($r = 0.092$), Q14 ($r = 0.383$), Q24 ($r = 0.395$), and Q30 ($r = 0.334$) were below the threshold of 0.40. Although Q24 and Q30 had been revised and provisionally retained after content review, their CITC values remained below the prespecified threshold during empirical item analysis. Therefore, Q7, Q14, Q24, and Q30 were removed, leaving 26 items for EFA. Please refer to Supplementary Table S3 for the complete Chinese and English wordings of final scale.

3.3 Exploratory factor analysis

The data were suitable for EFA, with a KMO value of 0.946 and a significant Bartlett's test of sphericity ($p < 0.001$). The five common factors extracted via EFA cumulatively accounted for 60.70% of the total variance. As shown in Table 3, the factor loading matrix indicated that the retained 26 items loaded clearly on the five hypothesized dimensions.

3.4 Confirmatory factor analysis and measurement invariance

CFA in the independent validation sample supported the correlated five-factor structure of the C-SSST-SF, with acceptable model fit, $\chi^2/df = 1.62$, CFI = 0.936, TLI = 0.928, RMSEA = 0.048, and SRMR = 0.050. The standardized CFA path diagram is presented in Figure 2, and the fit indices for the competing CFA models and measurement invariance models are summarized in Table 4. In contrast, the single-factor model fit the data poorly: CFI = 0.726 and RMSEA = 0.097. Although this comparison does not rule out all forms of common method bias, it suggests that a single general factor was unlikely to account for the observed item covariance structure.

Two additional alternative factor structures were examined. The higher-order model showed acceptable but slightly poorer fit than the correlated five-factor model ($\chi^2 = 493.26$, df = 294, $\chi^2/df = 1.68$, CFI = 0.923, TLI = 0.915, RMSEA = 0.050, SRMR = 0.057). The three-factor model showed clearly poorer fit ($\chi^2 = 713.55$, df = 296, $\chi^2/df = 2.41$, CFI = 0.838, TLI = 0.823, RMSEA = 0.072, SRMR = 0.069). These results supported retaining the correlated five-factor structure.

Across the continuing and dropout groups, the configural model provided acceptable fit. Constraining factor loadings to equality produced little change in fit (ΔCFI = −0.001), supporting metric invariance. Adding equality constraints on item intercepts also produced a small change in fit (ΔCFI = −0.004), supporting scalar invariance. These results indicate that group comparisons were not likely to be explained by measurement non-equivalence.

3.5 Reliability, convergent validity, and measurement error

As summarized in Table 5, all items exhibited significant standardized factor loadings on their respective latent variables, ranging from .596 to .881. The five dimensions demonstrated internal consistency, with Cronbach's $\alpha > 0.82$ and composite reliability (CR) >0.83. Furthermore, the average variance extracted (AVE) for each dimension exceeded the .50 threshold, indicating robust convergent validity. Regarding temporal stability and measurement precision, the intraclass correlation coefficients (ICCs) derived from the test-retest subsample ($n = 53$) ranged from 0.82 to 0.91, reflecting stability over time. The SEM values ranged from 0.28 to 0.34, while the MDC values ranged from 0.77 to 0.95 across the five dimensions.

3.6 Discriminant validity

Latent factor correlations ranged from .54 to .79. The Fornell–Larcker criterion was not fully satisfied for family support, whereas all HTMT values were below .85, providing complementary evidence that the constructs remained distinguishable under the HTMT criterion. In addition, the alternative model comparisons did not support replacing the correlated five-factor structure with either a higher-order structure or a model combining family, coach, and peer support into a single proximal interpersonal factor. Taken together, these findings supported retaining the five correlated but distinct factors, although the discriminant validity of family support should be interpreted cautiously (Table 6).

3.7 Known-groups comparisons

After scalar invariance was established, the continuing and dropout groups were compared on subscale and total scores. As detailed in Table 7, continuing participants reported higher support than participants in the dropout group across all five dimensions and on the total scale. The largest standardized difference was observed for family support ($d = 0.98$), followed by total support ($d = 0.95$). These unadjusted group differences should be interpreted cautiously because potential confounders were not controlled.

3.8 Supplementary classification evidence

Binary logistic regression and ROC analyses were conducted to examine whether C-SSST-SF scores were associated with school football participation status, with the main results for the total score summarized in Table 8. Group membership was coded as 0 = continuing and 1 = dropout. The total C-SSST-SF score was significantly associated with participation status, $B = -1.431$, SE = 0.182, OR = 0.239, 95% CI [0.167, 0.341], $p < 0.001$. This result indicated that higher perceived support was associated with lower

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TABLE 2 Item analysis and data normality test results for the exploratory sample.

Item	Mean	SD	Skewness	Kurtosis	CITC	*t*	Decision
Q1	4.02	1.05	−1.04	0.72	0.608	12.94	Keep
Q2	4.01	1.06	−1.07	0.73	0.589	12.21	Keep
Q3	3.90	1.07	−0.69	−0.35	0.592	12.33	Keep
Q4	3.99	1.10	−1.01	0.41	0.609	12.13	Keep
Q5	3.84	1.16	−0.84	−0.05	0.653	14.79	Keep
Q6	3.96	1.11	−1.05	0.44	0.569	10.90	Keep
Q7	3.17	1.29	0.02	−1.06	0.092	2.50	Delete
Q8	3.81	1.15	−0.69	−0.35	0.552	11.95	Keep
Q9	4.13	1.02	−1.06	0.49	0.706	15.22	Keep
Q10	3.14	1.31	−0.17	−1.07	0.445	8.29	Keep
Q11	4.07	0.98	−0.93	0.36	0.706	17.26	Keep
Q12	4.22	0.99	−1.50	2.07	0.486	9.56	Keep
Q13	4.17	1.08	−1.25	0.76	0.698	13.94	Keep
Q14	3.83	1.22	−0.79	−0.34	0.383	7.67	Delete
Q15	3.83	1.19	−0.82	−0.17	0.580	12.50	Keep
Q16	4.09	1.00	−0.98	0.44	0.526	8.78	Keep
Q17	4.02	1.03	−0.90	0.22	0.671	14.29	Keep
Q18	3.95	1.16	−0.99	0.15	0.675	15.08	Keep
Q19	4.16	0.93	−1.11	1.05	0.720	14.91	Keep
Q20	4.11	0.97	−1.06	0.85	0.708	14.12	Keep
Q21	3.97	1.07	−0.92	0.21	0.719	14.87	Keep
Q22	4.09	0.98	−1.04	0.71	0.757	15.99	Keep
Q23	4.22	0.93	−1.46	2.35	0.726	12.72	Keep
Q24	3.74	1.25	−0.72	−0.49	0.395	8.49	Delete
Q25	4.16	0.86	−0.99	1.19	0.706	13.88	Keep
Q26	4.22	0.83	−1.01	1.20	0.655	14.15	Keep
Q27	4.10	0.89	−0.80	0.34	0.715	14.77	Keep
Q28	4.05	0.89	−0.77	0.45	0.727	15.33	Keep
Q29	4.33	0.78	−1.24	1.90	0.654	12.14	Keep
Q30	4.06	1.10	−1.08	0.44	0.334	7.56	Delete

SD, standard deviation; CITC, corrected item-total correlation. The *t*-value represents the critical ratio derived from the extreme groups approach (comparing the upper 27% and lower 27% of total scores). All extreme-group comparisons were significant at $p < 0.001$, except for Q7, which was significant at $p < 0.05$.

odds of belonging to the dropout group. Specifically, each one-point increase in the total C-SSST-SF score was associated with 76.1% lower odds of belonging to the dropout group.

To provide dimension-level information without entering highly correlated subscales into the same model, additional single-predictor logistic regression models were conducted. As shown in Supplementary Table S4, all five subscale scores were negatively associated with dropout status, with ORs ranging from 0.339 to 0.452.

ROC analysis further showed that the total C-SSST-SF score distinguished continuing participants from dropout participants with an AUC of 0.756, 95% CI [0.708, 0.804], indicating acceptable overall discrimination. The optimal cut-off based on the Youden index was 3.654, with sensitivity = 0.590 and specificity = 0.792. However, the relatively low sensitivity indicates that approximately 41% of dropout participants would not be identified at this cut-off, limiting its suitability for individual-level screening. These findings should therefore be interpreted as supplementary classification evidence rather than evidence supporting use of the scale as a standalone screening tool.

4 Discussion

The present study provides initial psychometric evidence for the C-SSST-SF, a scale designed to assess perceived support among adolescents involved in Chinese school football. Previous studies have mainly focused on interpersonal sources of support and have paid less attention to institutional factors. The C-SSST-SF integrates interpersonal and institutional

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TABLE 3 Five-factor structure and factor loadings from exploratory factor analysis.

Item	Gov	School	Family	Coach	Peer
Q1	0.757				
Q2	0.634				
Q3	0.774				
Q4	0.595				
Q5	0.717				
Q6	0.604				
Q8		0.717			
Q9		0.532			
Q10		0.634			
Q11		0.641			
Q12		0.596			
Q13			0.583		
Q15			0.740		
Q16			0.828		
Q17			0.679		
Q18			0.729		
Q19				0.729	
Q20				0.732	
Q21				0.649	
Q22				0.879	
Q23				0.654	
Q25					0.826
Q26					0.743
Q27					0.640
Q28					0.651
Q29					0.815

KMO = 0.946. Cumulative variance explained = 60.70%. Extraction = principal axis factoring; rotation = Promax. Loadings below 0.30 are suppressed. Gov = governmental support.

support within the same measurement framework, thereby addressing this measurement gap. Research has found that the support adolescents receive is multifaceted, not simply a matter of interpersonal relationships. This type of support includes five related aspects: governmental support, school support, peer support, family support, and coach support (7, 15, 31). The reliability, measurement stability, and observed known-groups differences suggest that the C-SSST-SF may be useful for future research examining perceived support in school football.

One highlight of our research is that we considered social support within the broader context of school sports in China. Existing general and sport-specific instruments have assessed different interpersonal aspects of support, including perceived support from family and friends, support functions in sport, parental involvement, and the coach–athlete relationship (10–13). The inclusion of family, coach, and peer support in the C-SSST-SF is consistent with this prior literature. An important feature of the C-SSST-SF is the distinction between school and governmental support as two institutional sources. School support reflects organizational support directly experienced within the school, whereas governmental support reflects broader support that adolescents attribute to governmental or policy-related sources. In the present sample, these two dimensions were related but empirically distinguishable, providing support for their separate representation within the five-factor structure. At the same time, this psychometric distinction should not be interpreted as evidence that all adolescents necessarily draw an unequivocal cognitive boundary between school- and governmental-level support in every context. The governmental support dimension should therefore be understood as adolescents' perceived support attributed to governmental or policy-related sources, rather than as an objective indicator of governmental investment, policy provision, or implementation. This interpretation preserves the value of including a policy-related source of support while also recognizing the limits of what can be inferred from the present psychometric evidence. Although family, coach, and peer support were relatively highly correlated, they represent support from distinct social agents. The alternative model comparisons further showed that the higher-order model had acceptable but somewhat poorer fit than the correlated five-factor model, whereas combining family, coach, and peer support into a single proximal interpersonal factor resulted in clearly poorer fit. These findings therefore support retaining the five correlated but distinct support dimensions, although discriminant validity should be interpreted cautiously for family-related pairings.

C-SSST-SF scores differed between the continuing and dropout groups across all five support dimensions. Family support showed the largest standardized difference (Cohen's $d = 0.98$). This finding is consistent with research describing parents as important "gatekeepers" of youth sport participation (26, 58, 59). During the transition from primary to junior secondary school, when academic demands often increase, family support may play an important role in whether adolescents are able to continue participating in school football (26, 60).

Moreover, the standardized differences for school and governmental support were larger than those for coach and peer support. These findings suggest that institutional sources of support may also be relevant in this educational context. From a practical perspective, the C-SSST-SF may help schools, coaches, and education administrators identify domains in which adolescents report relatively lower interpersonal or institutional support during the transition period. However, the present cross-sectional comparisons cannot establish whether these support differences causally influence dropout (24). More broadly, the ecological systems framework underlying the C-SSST-SF may also be relevant to other school-based sports in which participation is shaped by both interpersonal and institutional sources of support. However, application beyond football would require sport-specific adaptation and further psychometric validation.

The supplementary logistic regression and ROC analyses further strengthened the known-groups evidence. Higher total C-SSST-SF scores were associated with lower odds of belonging to the dropout group, and the total score showed acceptable discrimination between continuing and dropout participants. However, the sensitivity of 0.590 limits the scale's current utility for individual-level classification. These findings should therefore be interpreted as supplementary classification evidence rather than criterion-related or predictive validity evidence.

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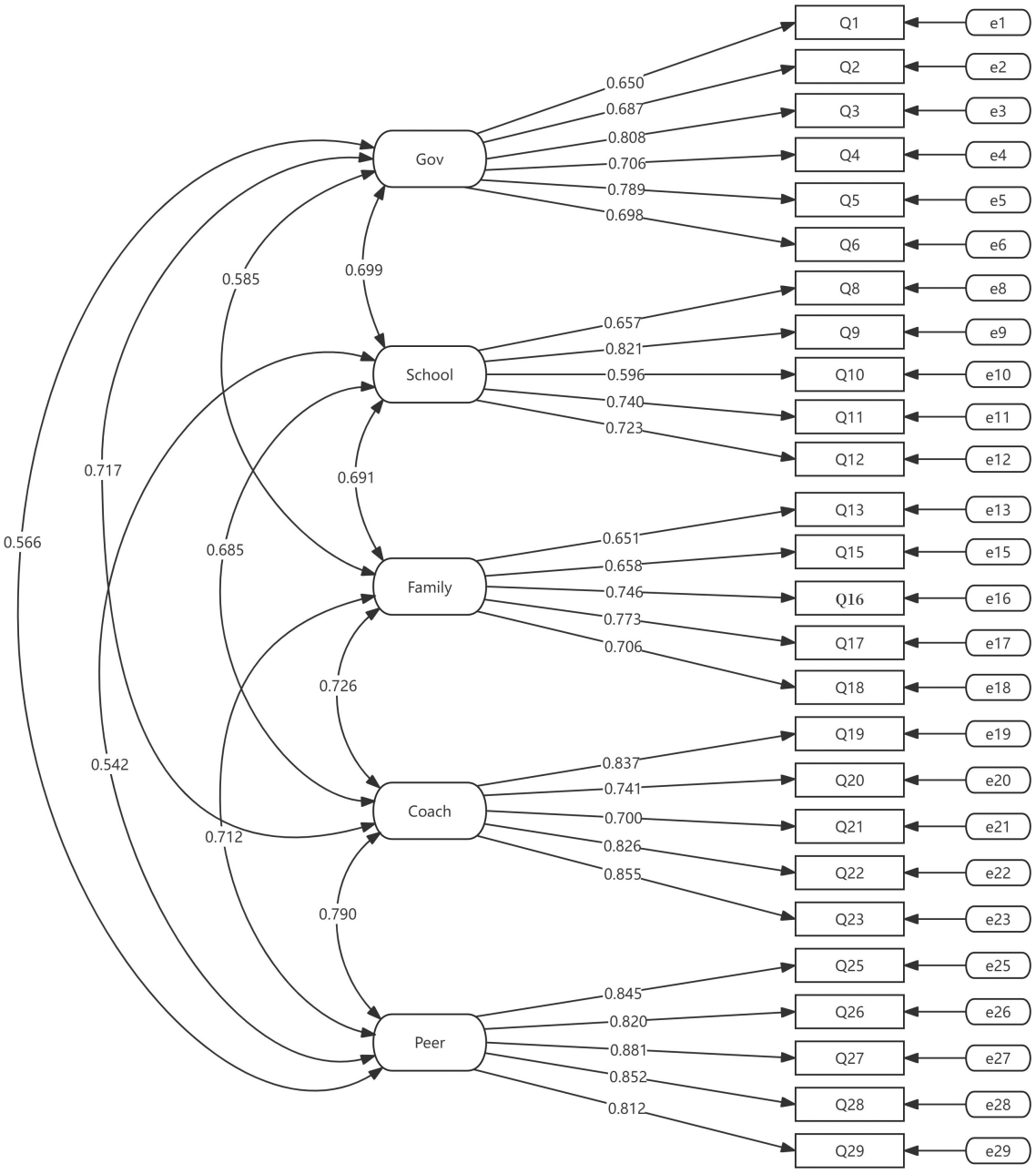


FIGURE 2
CFA path diagram of the C-SSST-SF. Standardized factor loadings are shown on the single-headed paths, and latent factor correlations are shown using curved double-headed arrows. All factor loadings were statistically significant at $p < 0.001$. Latent factor correlations are also reported in Table 6. Gov = governmental support.

A major strength of this study lies in its split-sample cross-validation design and its use of COSMIN-informed measurement properties, including internal consistency (44), test–retest reliability, structural validity, measurement invariance, and absolute measurement error indices. Nevertheless, several limitations should be noted. First, criterion-related validity was not directly examined because no established external measure was included. Although the present study provided known-groups and supplementary classification evidence by comparing continuing and dropout participants, future research should examine the associations of C-SSST-SF scores with established sport social support measures, sport motivation, basic psychological need satisfaction, continuation intention, dropout intention, and prospective dropout outcomes. In addition, the known-groups comparisons did not control for potentially relevant factors such as years of training, competition level, academic status, or minor injury history. Therefore, the observed differences in perceived support may partly reflect unmeasured confounding and should be interpreted cautiously. Second, the sample was geographically imbalanced, with 93.2% of participants recruited from Central China and only two participants from Western China. Therefore, the present psychometric findings should not be interpreted as nationally representative of adolescents

TABLE 4 Model fit indices for CFA models and measurement invariance testing.

Models	χ^2	Df	χ^2/df	CFI	TLI	RMSEA	SRMR	ΔCFI	ΔRMSEA
Competing CFA models									
Model 1: Single-factor model	1067.69	299	3.57	0.726	0.702	0.097	0.088	–	–
Model 2: Correlated five-factor model	469.29	289	1.62	0.936	0.928	0.048	0.050	–	–
Model 3: Higher-order model	493.26	294	1.68	0.923	0.915	0.050	0.057	–	–
Model 4: Three-factor model	713.55	296	2.41	0.838	0.823	0.072	0.069	–	–
Measurement invariance models									
Model 5: Configural invariance	942.75	578	1.63	0.936	0.928	0.048	0.047	–	–
Model 6: Metric invariance	969.00	599	1.62	0.935	0.929	0.048	0.051	−0.001	0.000
Model 7: Scalar invariance	1012.61	620	1.63	0.931	0.928	0.048	0.053	−0.004	0.000

$N = 541$ for measurement invariance models; $n = 271$ for competing CFA models. The higher-order model specified the five first-order support dimensions under a general social support factor. In the three-factor model, the three latent dimensions were governmental support, school support, and proximal interpersonal support, with family, coach, and peer support combined into the proximal interpersonal support factor. χ^2, chi-square; df, degrees of freedom; CFI, comparative fit index; TLI, Tucker–Lewis index; RMSEA, root mean square error of approximation; SRMR, standardized root mean square residual. ΔCFI and ΔRMSEA refer to changes from the previous, less constrained model. All χ^2 values were significant at $p < 0.001$.

TABLE 5 Reliability, convergent validity, and measurement error indices.

Construct	Items	Loadings	α	CR	AVE	ICC (95% CI)	SEM	MDC
Governmental support	6	.650–.808	0.868	0.869	0.526	0.91 (0.84–0.95)	0.28	0.77
School support	5	.596–.821	0.828	0.835	0.506	0.89 (0.82–0.94)	0.28	0.77
Family support	5	.651–.773	0.832	0.834	0.502	0.87 (0.74–0.93)	0.33	0.93
Coach support	5	.700–.855	0.889	0.895	0.630	0.87 (0.77–0.93)	0.34	0.95
Peer support	5	.812–.881	0.923	0.924	0.709	0.82 (0.70–0.90)	0.34	0.93

$n = 271$ for CFA, internal consistency, CR, and AVE; $n = 53$ for test-retest reliability. All standardized factor loadings were significant at $p < 0.001$. α, Cronbach's alpha; CR, composite reliability; AVE, average variance extracted; ICC, intraclass correlation coefficient; SEM, standard error of measurement; MDC, minimal detectable change.

TABLE 6 Latent factor correlations, HTMT ratios, and discriminant validity.

Dimension	1.Gov	2.School	3.Family	4.Coach	5.Peer
1.Gov	(0.725)	0.682	0.587	0.717	0.566
2.School	0.699*	(0.711)	0.688	0.678	0.540
3.Family	0.585*	0.691*	(0.709)	0.728	0.713
4.Coach	0.717*	0.685*	0.726*	(0.794)	0.790
5.Peer	0.566*	0.542*	0.712*	0.790*	(0.842)

Values on the diagonal are the square roots of AVE. Values below the diagonal are latent factor correlations from the CFA. Values above the diagonal are HTMT ratios. The Fornell–Larcker criterion was not fully satisfied for family support, whereas all HTMT values remained below .85.
*$p < 0.01$.

TABLE 7 Mean comparison of C-SSST-SF scores between the continuing and dropout groups.

Dimension	Continuing group ($n = 419$)	SD	Dropout group ($n = 122$)	SD	t	Cohen's d
	M		M			
Governmental support	4.10	0.80	3.49	0.79	7.45*	0.77
School support	4.00	0.79	3.40	0.76	7.47*	0.77
Family support	4.18	0.74	3.41	0.91	9.51*	0.98
Coach support	4.25	0.75	3.72	0.84	6.69*	0.69
Peer support	4.29	0.68	3.84	0.75	6.16*	0.63
Total Scale	4.16	0.62	3.57	0.61	9.26*	0.95

Continuing group ($n = 419$); dropout group ($n = 122$). SD, standard deviation. Cohen's d indicates the standardized mean difference effect size.
*All t-tests are significant at $p < 0.001$.

TABLE 8 Logistic regression and ROC analysis for distinguishing dropout status.

Panel A. Logistic regression					
Predictor	*B*	SE	OR	95% CI	*p*
Total C-SSST-SF score	−1.431	0.182	0.239	0.167–0.341	<.001
Panel B. ROC analysis					
Predictor	AUC	95% CI	Cut-off	Sensitivity	Specificity
Total C-SSST-SF score	0.756	0.708–0.804	≤3.654	0.590	0.792

Group membership was coded as 0 = continuing and 1 = dropout. OR = odds ratio; AUC = area under the receiver operating characteristic curve. The cut-off was determined using the Youden index and should be interpreted as an exploratory, sample-specific classification threshold for distinguishing continuing and dropout status, rather than a prospective dropout-risk threshold. These analyses were interpreted as supplementary classification evidence rather than criterion-related validity evidence.

participating in school football across China. Future studies should validate the C-SSST-SF in more geographically diverse samples, including eastern, western, urban, and rural regions, to examine whether the factor structure and the relative importance of governmental support remain stable across different school-football contexts. Third, girls accounted for only 24.9% of the valid sample, and neither sex-based differential item functioning (DIF) nor cross-gender measurement invariance was examined. Given that girls' support needs in youth sport may differ from those of boys (61, 62), the psychometric evidence for female adolescents remains less established, and the scale should be interpreted cautiously when used with female school-football participants. Future studies should recruit larger female samples and further examine sex-based DIF and measurement invariance. Fourth, the cross-sectional design limits conclusions about predictive validity (4). Longitudinal studies are needed to examine whether baseline C-SSST-SF scores prospectively predict later continuation, reduced participation, or dropout during the transition from primary to junior secondary school.

5 Conclusion

Our study independently developed the C-SSST-SF, which was determined to have a 26-item, five-factor structure and demonstrated satisfactory psychometric properties in the present sample.

Supplementary logistic regression and ROC analyses indicated that C-SSST-SF scores were associated with school-football participation status and provided preliminary classification evidence. However, the sensitivity of the ROC-derived cut-off was limited, and the scale should not currently be used as a standalone screening tool for individual dropout risk. The C-SSST-SF may instead serve as a context-sensitive measure for assessing perceived interpersonal and institutional support in future research on Chinese school football.

Data availability statement

The data that support the findings of this study are not publicly available due to privacy and ethical restrictions regarding the underage participants. However, de-identified data are available from the corresponding author upon reasonable request and with permission from the Institutional Review Board.

Ethics statement

The study involving human participants was approved by the Biomedical Research Ethics Committee of Hunan Normal University (Approval Number: 2026–306). Written informed consent was obtained from parents or legal guardians, and assent was obtained from adolescent participants before data collection.

Author contributions

JH: Data curation, Writing – original draft, Writing – review & editing, Conceptualization, Investigation, Software, Validation. HL: Software, Supervision, Writing – review & editing. YC: Project administration, Resources, Writing – review & editing. WZ: Investigation, Visualization, Writing – review & editing. ZY: Data curation, Writing – review & editing. BY: Formal analysis, Writing – review & editing. LY: Project administration, Validation, Supervision, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fped.2026.1897283/full#supplementary-material

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